

A researcher's walkthrough

Follow this end to end and you will have used every part of Research Companion.

You do not need to know the tool, the CLI, or anything about the topic. Every step says what to type or click, what you get back, what the tool is doing underneath, and how to decide what to do next.

The running example is a researcher starting cold on **LM inference** — no papers, no draft, no idea what the open problems are.

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Before you start

```
pip install "research-companion[server]"
```

You need **one** model key — Anthropic or OpenAI. Put it in a `.env` beside where you run the tool (copy `.env.example`), or paste it into Settings later:

```
cp .env.example .env      # then fill in ANTHROPIC_API_KEY or OPENAI_API_KEY
```

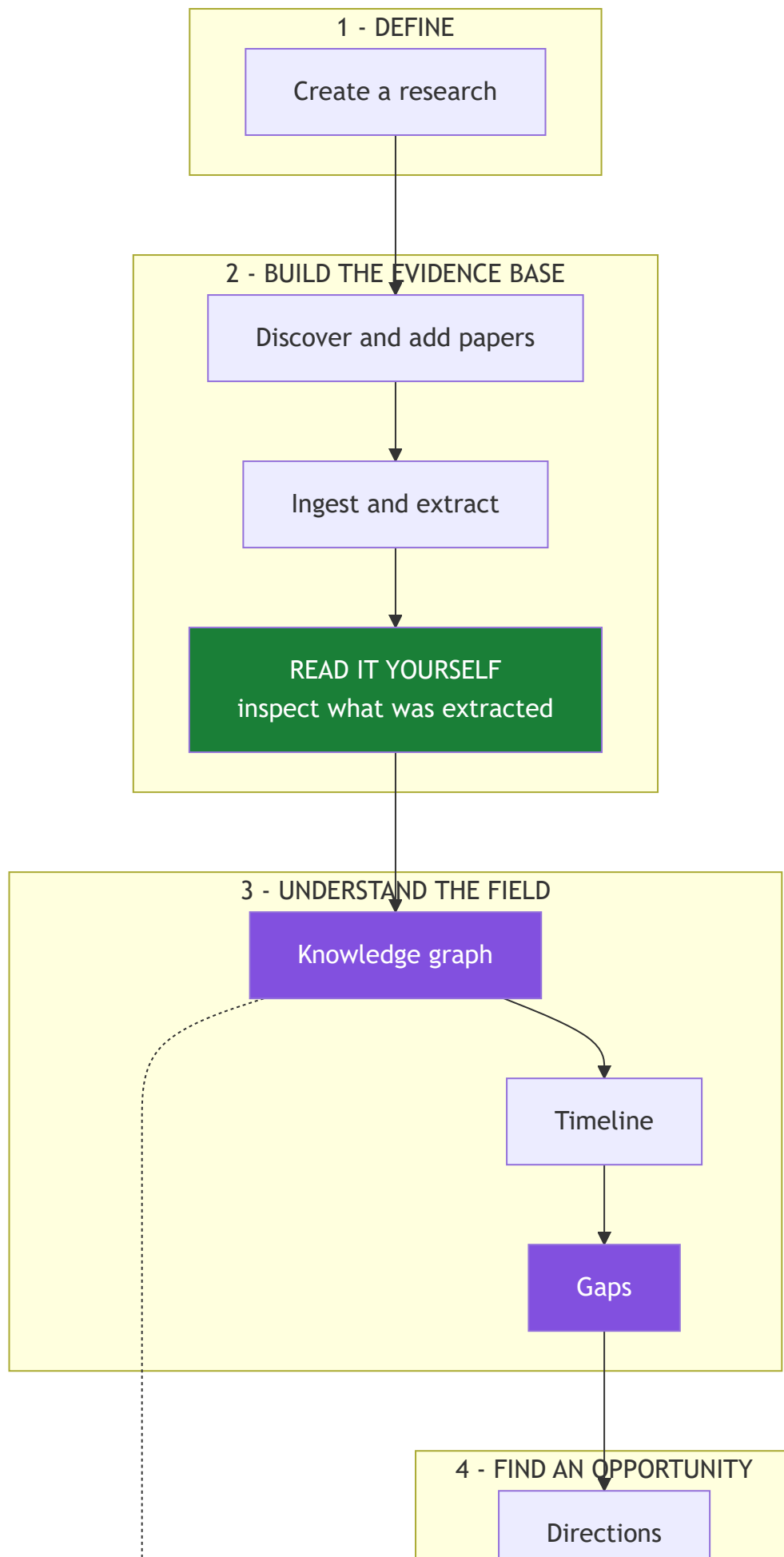
Start the browser Lab:

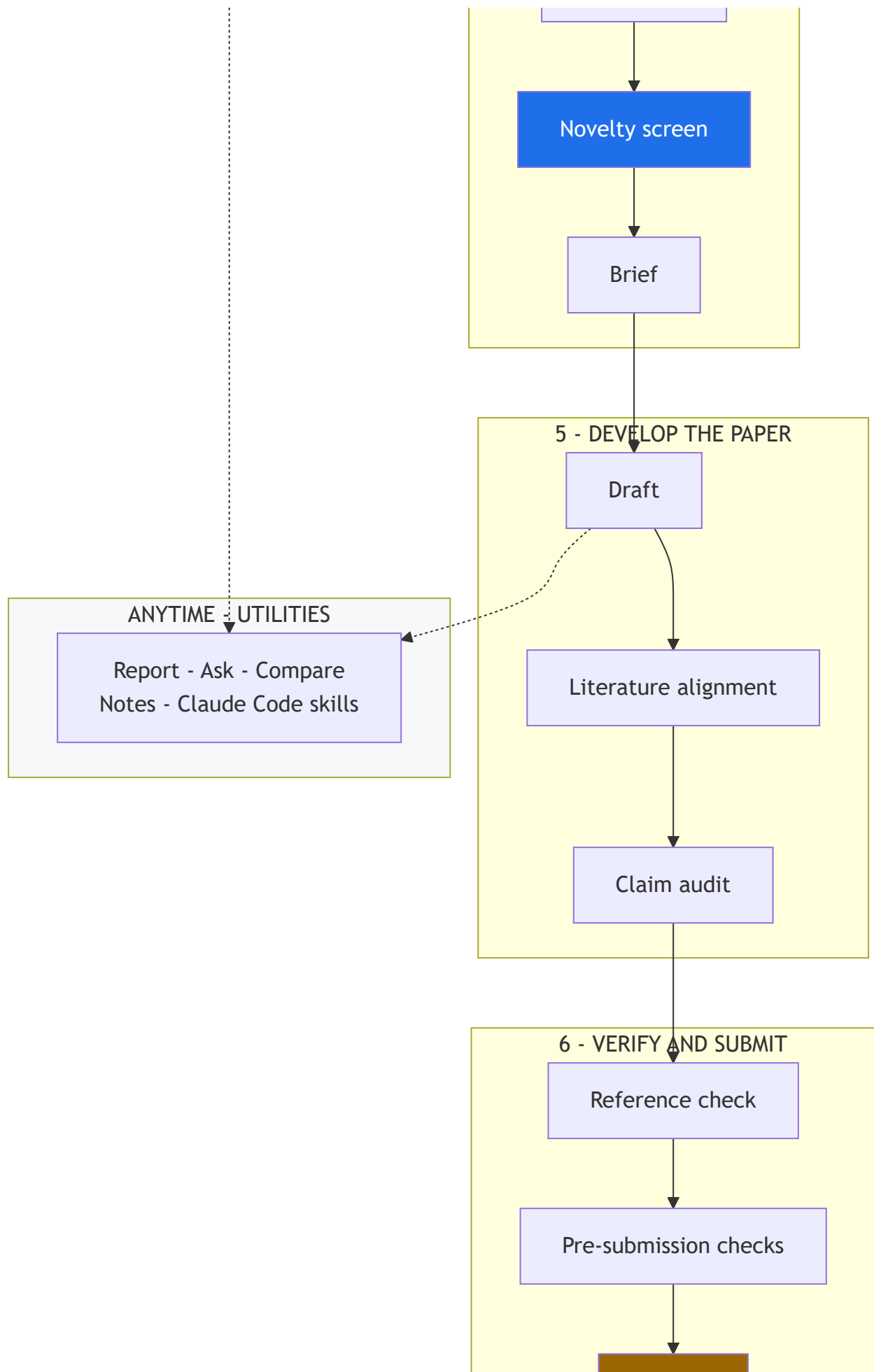
```
research-companion lab serve
```

Open **<http://127.0.0.1:8800>**. Everything below happens there, and every step also shows the terminal equivalent if you prefer the CLI.

Your papers stay on your machine except when a step explicitly calls a model or a public catalogue. The deterministic half of the tool — search, graph, timeline, reference checking, statistics checking — runs with no key at all.

The whole journey in one picture





Three things to notice before you begin.

The order is the point. You understand the field *before* you pick a direction — graph, then timeline, then what the authors say is unfinished. Only then do directions and the novelty screen make sense, because by then you can judge them. A tool that proposed directions first would be asking you to trust it blindly.

You read before the machine reasons. Step 5 sits deliberately between ingestion and everything derived from it. What gets extracted from your papers becomes the input to the graph, the gaps and every direction — so look at it while it is still cheap to correct.

Nothing is enforced. These are tabs, not a wizard. Stop after Phase 3 and you have a searchable, cited library. Skip Phases 1–4 entirely and start at Step 12 with a draft you already wrote. The order is advice, not a gate.

Phase 1 — Define the research

Step 1 — Create a research

A **research** is an isolated workspace: its own papers, its own graph, its own draft. Keeping projects apart matters, because a graph mixing LM inference papers with unrelated ones produces worse directions and noisier gaps.

Do this. Open the Lab. On a fresh install you land on a two-path chooser — pick **Brainstorm from an idea**. The top bar reads `Research: none` until you name one; that is honest, not broken.

In **Brainstorm**, type your topic and click **Name your research**:

```
LM inference
```

Terminal equivalent:

```
research-companion workspace new "LM inference"  
research-companion workspace use "LM inference"
```

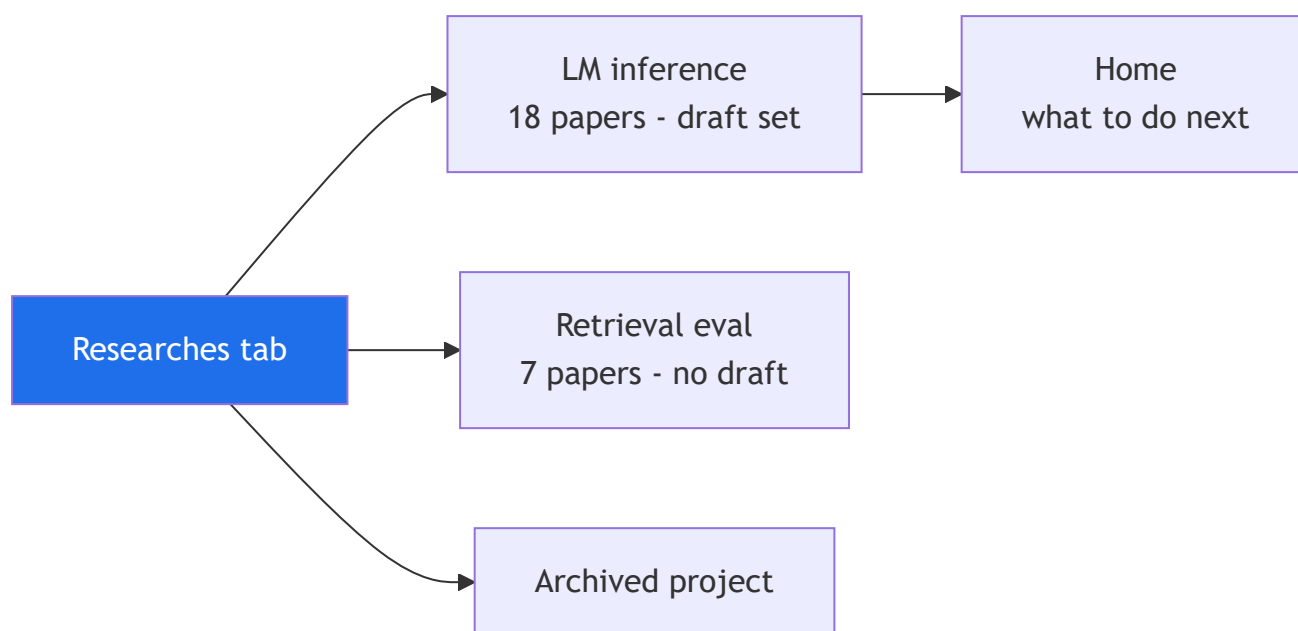
What you get. An empty research, now active in the top bar. Every later step writes here and nowhere else.

Step 2 — Home and Researches: knowing where you are

Two tabs exist to answer "what now?" and "what else am I working on?".

Home is the dashboard. On an empty research it offers the two-path chooser. Once you have papers it becomes a journey view: what you have done, what the obvious next step is, and a timeline of recent activity. If you come back after a fortnight and cannot remember where you left off, start here.

Researches is the table of every project you have — papers, how many are analysed, how many failed, whether a draft is set and how many versions, citation coverage, the mix of supporting versus contradicting evidence, open items, and when each was last touched. Sortable, with rename / archive / delete and a + **New research** button.



Keep one research per project. It is the single most effective thing you can do for the quality of directions, gaps and reports, because every one of them reads the whole library.

Phase 2 — Build the evidence base

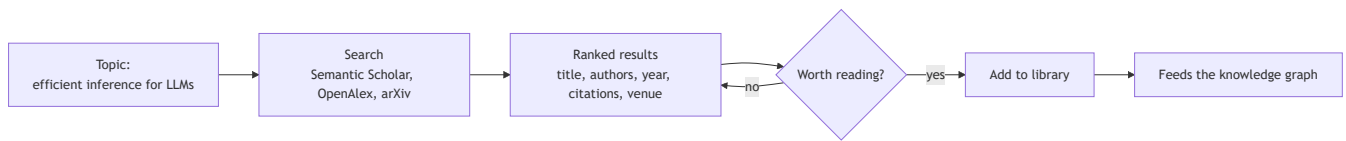
Step 3 — Discover papers and build the library

You have a topic and nothing else. This step turns it into papers.

Do this. In **Brainstorm**, type a topic and click **Search**:

efficient inference for large language models

What you get. Real papers from the literature — title, authors, year, citation count, venue. Papers already in your library are marked, so you never add the same one twice.



How to choose, as a newcomer. You are not trying to be exhaustive. Aim for **10–20 papers that disagree with each other** — a survey, two or three highly-cited method papers, and something from the last year. A library of near-identical papers produces directions that all say the same thing.

For LM inference that might mean KV-cache compression, speculative decoding, quantisation, batching and serving systems. That spread is what later lets the tool show you where the field is crowded and where it is thin.

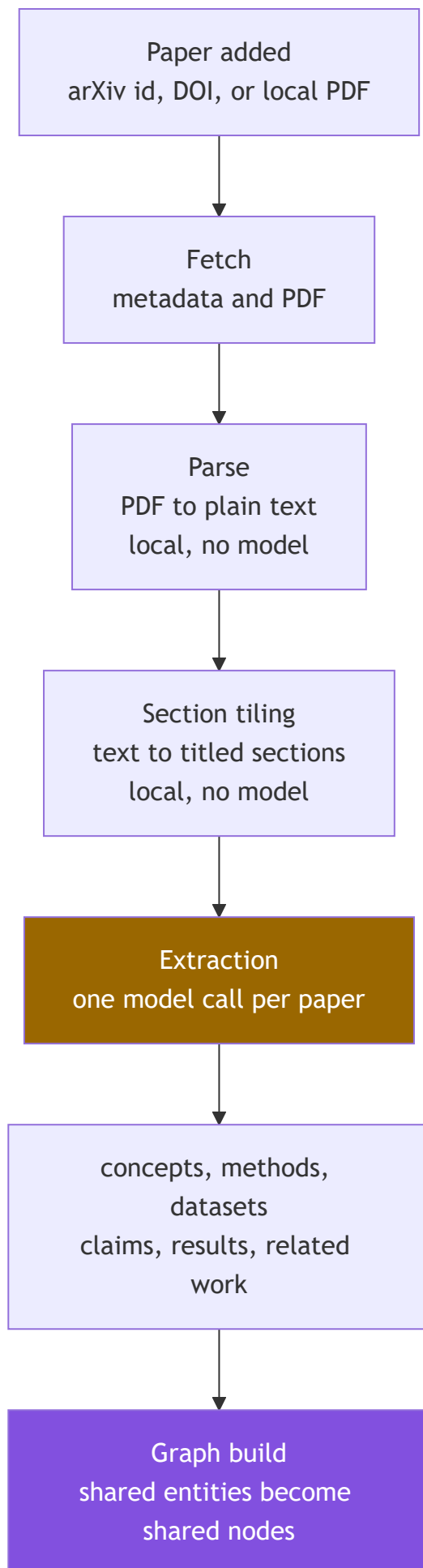
Terminal equivalent:

```
research-companion discover "efficient inference for large language models"
research-companion add 2211.17192          # arXiv id, DOI, S2 id, or a local PDF
```

Why searching is free. Catalogue search is a public API call, not a model call. You only start spending when a paper is analysed in Step 3.

Step 4 — What happens when a paper is added

This is the only step that is mostly invisible, so here is exactly what runs.



What you get. Each paper becomes a structured record: what it claims, what methods it uses, which datasets, what it reports, and what it cites. That record is what lets later answers cite down to the *section*, not just the paper.

Watch for. Ingestion is asynchronous — keep working while it runs; the Activity indicator shows progress. A scanned PDF with no text layer fails loudly rather than silently producing an empty paper.

Step 5 — Read it yourself, before the machine reasons over it

Everything after this point is computed from what Step 4 extracted: the graph, the gaps, the directions, the report. If an extraction is wrong, that error propagates silently into every conclusion downstream and looks like analysis. **This is the cheapest moment to catch it.**

Do this. Open **Library**, click any paper. Read what was pulled out of it — the claims, methods and datasets — against the actual paper.

What you get. A reader with the original PDF and a text view, plus a **Simplified** plain-English view for papers outside your area — useful when you are new to a field and the notation is the obstacle.

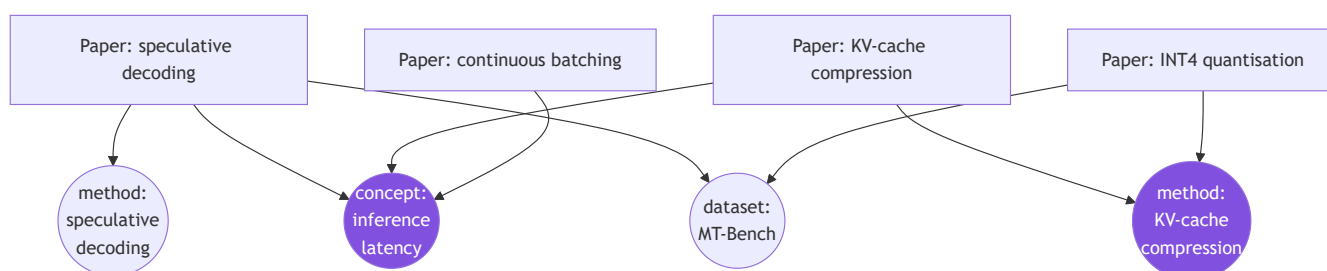
Save anything you find with **Save note**: from an alignment card, an opportunity, the reader, or a brief bullet. The **Notes** tab collects them into one filterable list you can export as a revision checklist.

Phase 3 — Understand the field

Step 6 — The knowledge graph

Do this. Open the **Graph** tab.

What you get. A concept-level map of your library — not a citation network. Nodes are the *ideas*: concepts, methods, datasets, claims. If five papers use KV-cache compression, that is **one node linked to all five**.



What this tells you at a glance — and what it does not.

Everything here is **density within your corpus**, not within the field. Load six quantisation papers and one on speculative decoding, and quantisation will look dominant no matter what the literature actually looks like. Read every observation below as "...among the 18 papers in this research."

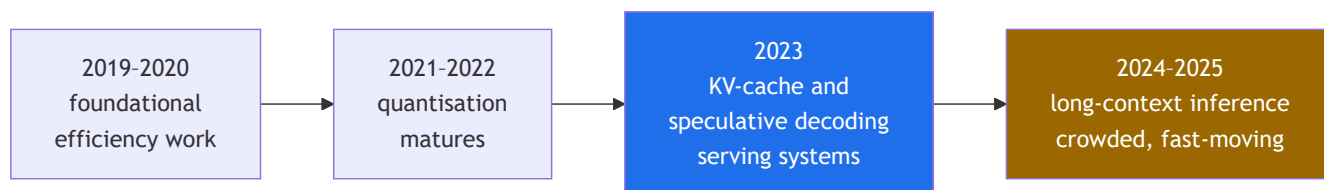
- A node touched by **many of your papers** is dense in your corpus. That is a hint the area is crowded, worth confirming against the Timeline and a wider search.
- A node touched by **one paper** is either a dead end, an opening, or simply a gap in *your reading*. Step 8 (Gaps) helps you tell which.
- Two clusters that **share no nodes** are two conversations that have not met in your corpus — often where the interesting work is, and also the first thing to check you have not just under-read one of them.

Click a node to see every paper that touches it. Click a paper to open it in the reader at the exact section.

Step 7 — Timeline: how the field moved

Do this. Open **Timeline**.

What you get. Your library laid out by publication year — when a method first appears, which years are crowded, where the recent work sits.



Why a newcomer should look here early. It tells you whether the direction you picked in Step 5 joins a *new* conversation or a *long-running* one. Both are fine; they need very different framing in a paper.

Step 8 — Gaps: what the papers admit is unfinished

Do this. Open **Gaps**.

What you get. Every paper's own stated limitations and future work, collected and grouped into **themes across your whole library** — deduplicated, ranked and citation-backed, typed (limitation vs future work), classified by kind (method, resources, evaluation, application, problem), and flagged open / partial / addressed.

This is the most under-used feature for a newcomer. It is not the tool guessing where the gaps are — it is **the authors themselves** saying what they could not do.

Read a repeated theme as a **recurring stated gap**, not a confirmed open problem. Six papers can all name a limitation that a seventh — one you have not added — has since solved. That is why each theme carries a status:

status	meaning
open	nothing in your library addresses it
partially	something addresses part of it
addressed	a paper in your library claims to solve it

An `addressed` theme whose supporting quote could not be verified is downgraded to `partially` automatically — the tool will not upgrade a gap to solved on evidence it could not confirm.

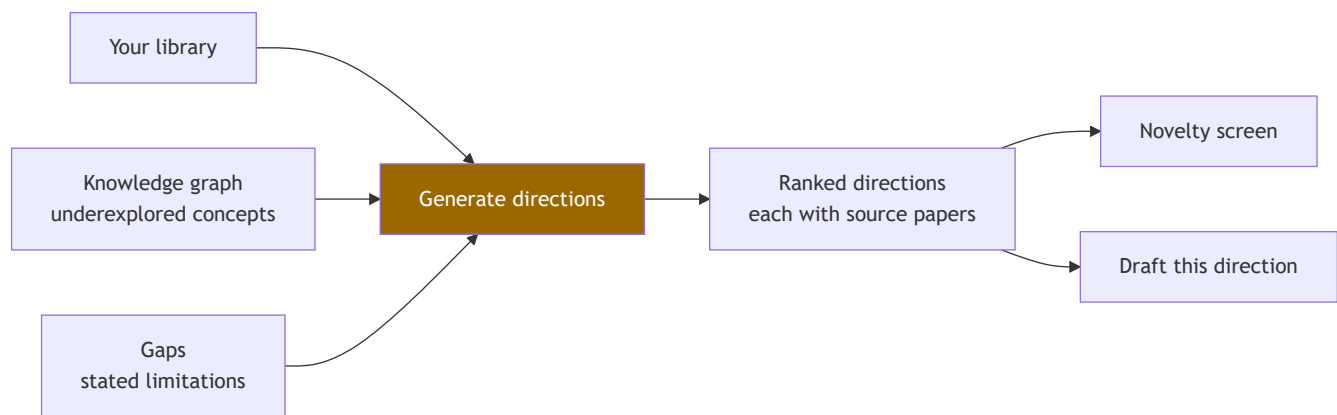
Phase 4 — Find an opportunity

Step 9 — Research directions

You have a library and a graph. This step proposes what you could actually work on.

Do this. In **Brainstorm**, click **Generate directions**.

What you get. A ranked list of candidate directions, each carrying the papers it came from — so you can check the reasoning rather than trust it.



A direction is a *hypothesis about where the work is*, generated by a model from your library. It is a starting point for your judgement, not a verdict — the citations under each one exist so you can disagree with it.

Step 10 — Novelty screen

Do this. On a direction you like, click **Check novelty**.

What you get. The closest prior work it could find, and how close it is. This is the one reasoning step that deliberately looks **outside your library** — it searches the public catalogues (Semantic Scholar, OpenAlex, arXiv), because checking novelty against only the papers you already chose would be circular.

The outcome is graded, not a yes/no:

outcome	meaning
novel	nothing close found in the sources searched
incremental	related work exists; this extends it
overlaps	prior work covers part of this
anticipated	prior work already does substantially this

Read it as a screen, not a verdict. "Novel" means nothing close was found **in the sources searched** — never that nothing exists. If the search returns no prior work at all, the tool says so explicitly and makes no model call, because there is nothing to compare against. Judge it together with what it screened, not alone.

Step 11 — The brief

Do this. Click **Generate brief**.

What you get. Section headings, each seeded with **cited bullet points** drawn from your papers — the raw material of a related-work section. Every bullet links back to its source, and you can attach your own notes to any of them.

This is the bridge between reading and writing: instead of a blank page you start editing something that already has citations attached.

Your whole Brainstorm session — topic, papers, directions, brief — is saved per research. Close the tab, come back tomorrow, it is still there.

Phase 5 — Develop the paper

Step 12 — Turn a direction into a draft

Do this. On the direction you chose, click **Draft this direction**.

What you get. A real, editable draft outline in the **Draft** tab, structured into sections, ready to flow into everything below.

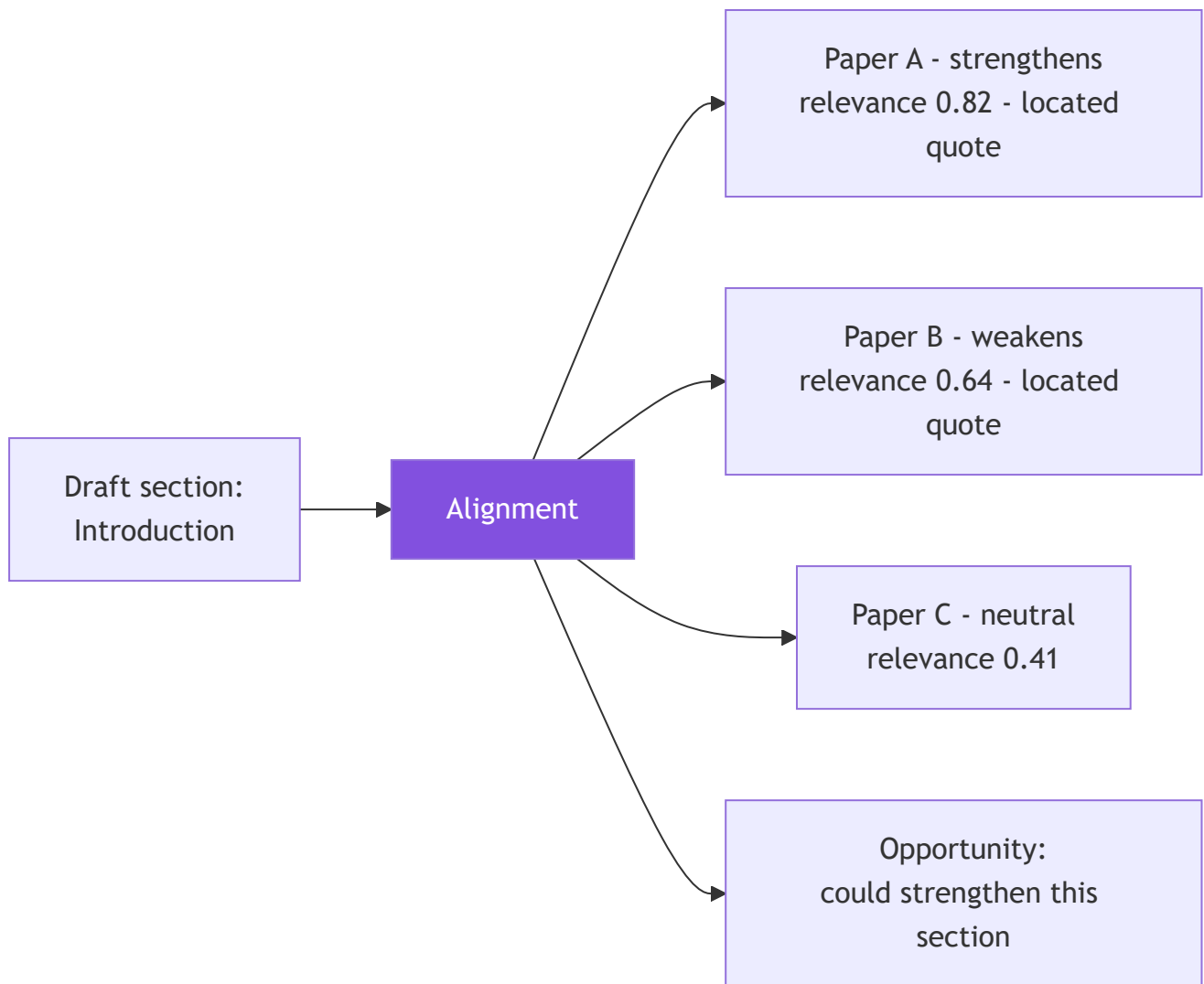
Already have a draft? Skip all of the above — put your PDF or text in the Draft tab and set it as the draft:

```
research-companion set-draft <paper_id>
```

Step 13 — Analyse your draft against the literature

Do this. Open **Draft**, click **Analyze this draft**.

What you get. For every section of your draft, the papers that relate to it — each with a **stance** (strengthens / weakens / neutral), a relevance score, a rationale, and a **verified quote** from the source.



What "verified" means here, precisely. The quote was checked to appear **verbatim** in the source, and carries a locator that can find it again. That is a *text match* — it is not a check that the source supports the point being made around it. Those are different questions, and Step 14 is the one that asks the second. A quote that cannot be located is shown as unverified rather than quietly dropped.

And the percentages are relevance, not confidence. 82% is a retrieval relevance score for how strongly the passage matched the section — not a probability that the paper supports you. The Report tab states this on screen: relevance and stance are AI judgements, not independently verified.

Step 14 — Are the cited sources actually saying that?

Existing checks answer two easier questions: does the reference *exist*, and does a quote appear *verbatim*. Neither catches the failure that matters most: a real citation attached to a claim the source never makes.

Do this. Turn on **claim_audit** in Settings, then click **Check citations** in Draft or Report.

What you get. A verdict per cited passage:

verdict	meaning
Supported	the passage supports the claim
Not supported	it does not — <i>the only adverse verdict</i>
Could not check	passage unavailable, or the model was unsure
No anchor	the citation has no locator to check against
Not checked	the audit did not run

The last three are **failures to check**, not findings against the citation, and are shown that way. The auditor is deliberately biased toward "unclear": at the rate real miscitation occurs, a noisy checker's warnings would be mostly wrong, and people would learn to ignore all of them.

> **Costs one model call per cited passage. That is why it is opt-in.**

Phase 6 — Verify and submit

Step 15 — Citations: does your bibliography hold up?

Do this. Open **Citations**, or run:

```
research-companion refcheck <paper_id>
```

What you get. Every reference looked up in CrossRef, OpenAlex and arXiv — with **no model calls at all**.

verdict	meaning
verified	a matching record was found and the details agree
suspect	a record was found but something disagrees — often a typo, or a preprint/published mismatch
not found	no matching record in the catalogues searched

"Not found" is not "fabricated." Books, theses, workshop papers, technical reports, very recent preprints and non-English venues land there routinely. Treat it as *check this one by hand*. For biomedical work add `--connectors europepmc`, or PubMed-only references will look missing for lack of coverage.

The output also reports lines it **could not parse** as references. Those were never checked, and they are counted separately so the totals cannot be mistaken for the whole bibliography.

Step 16 — Pre-submission checks

All three are deterministic and cost nothing.

```
research-companion check-compliance <paper_id> --venue neurips
research-companion check-stats      <paper_id>
research-companion check-overlap    <paper_id>
```

check	catches
compliance	page limit, abstract length, missing required sections (limitations, broader impact, ethics) — the desk-reject list
statcheck / GRIM	reported p-values that do not match their test statistic; means impossible for the stated sample size
overlap	passages near-duplicating another paper in your own library — self-plagiarism, dual submission

Supported venues: `neurips` `icml` `iclr` `aaai` `acl` `emnlp` `cvpr` `kdd` `sigir` `nature` `science` `pnas`
`nature-medicine` `nature-physics` `the-lancet` `bioinformatics` `prl` `psych-science` `aer`

0 statistical tests found means nothing was checked — not that your statistics are sound. The tool separates "clean" from "not checked" everywhere, and so should you when reading it.

Step 17 — Export

```
research-companion export-bib --format bibtex > refs.bib
research-companion cite-tex <paper_id>                # a LaTeX cite key
```

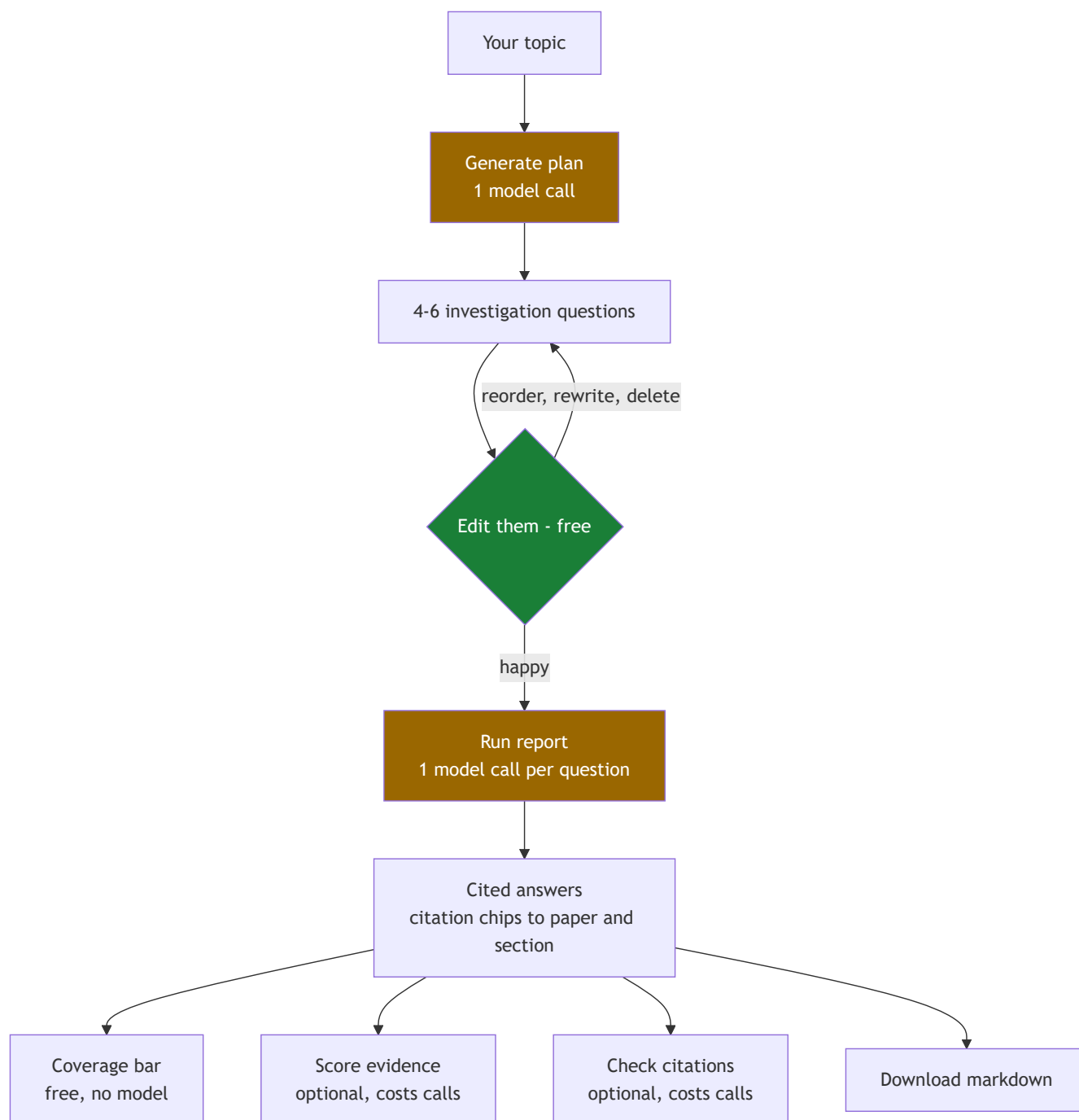
The Report tab has **Download (.md)** — the whole review as markdown, with its citations, badges and honesty caveats intact rather than stripped.

Notes export as a revision checklist.

Anytime — utilities, not stages

Report: a cited literature review

Do this. Open **Report**. Type a topic and click **Generate plan** first.



Why plan first. Planning is one call and editing is free; answering costs one call per question. Reviewing the questions before paying for answers is the cheap checkpoint — and it is usually where you notice the tool misread your topic.

What you get. A structured review of *your library* — never the open web — where every claim carries a citation chip back to the exact paper and section.

Reading the extras honestly:

signal	what it means
Coverage	how much of the material <i>our own keyword search</i> judged relevant actually got cited. A BM25 heuristic, not ground truth.
Score evidence	an AI judgement of each citation's relevance and stance (supports / contradicts / neutral). Labelled a judgement, never "verified".
Check citations	see Step 15.

Ask and Compare

Ask — a question in plain language, answered from your papers with a citation after every claim. If your library does not cover it, it says so instead of guessing. That refusal is the feature.

Compare — two papers side by side: what each claims, methods, datasets, and where they agree or diverge. Useful when two papers report conflicting numbers and you need to see why.

```
research-companion ask "why does KV-cache compression hurt long-context quality?"
research-companion compare <paper_a> <paper_b>
```

From inside Claude Code

The free checks also run as [Claude Code](#) skills, so you can ask in plain language instead of learning flags:

```
cp -r skills/refcheck skills/submission-check ~/.claude/skills/
```

```
/refcheck paper.pdf
/submission-check paper.pdf --venue neurips
```

They also fire on the question phrased naturally — "*are these citations real?*", "*will this get desk-rejected?*" Both write to a scratch workspace, so they can never touch your real research.

For other agents, `research-companion mcp serve` exposes the same deterministic verification tools over MCP.

What it costs

free — no model call	costs model calls
literature search and discovery	paper extraction (once per paper)
PDF parsing, section tiling	research directions
the knowledge graph view	novelty check
timeline	brief
reference checking (<code>refcheck</code>)	report questions and answers
statcheck / GRIM	evidence scoring
venue compliance	claim audit (one call per passage)
overlap detection	draft alignment
coverage bars	Ask, Compare, Chat
notes, export, BibTeX	

Before a big build:

```
research-companion cost-estimate
```

What it will never do

Worth knowing before you trust any of it.

- **It does not answer from the model's memory.** Every claim it makes is traced to a document — your library for anything about *your* papers, and public catalogues (Semantic Scholar, OpenAlex, arXiv, CrossRef) for discovery, novelty screening and reference checking. Nothing is asserted because a model recalled it.
- **It does not silently widen its scope.** Ask, Compare, Report, Gaps and draft alignment read *only* your library. If you did not add it, it does not exist to those features.
- **It does not judge whether your idea is good.** Directions and novelty checks are grounded suggestions, not verdicts.
- **It does not hide what it could not do.** A check that did not run is never shown as a check that passed — "could not check" and "clean" stay separate.
- **It does not call a missing reference fake.** Not found means not found in the catalogues searched.
- **It does not spend money without asking.** Every costed action is a button you press, priced before you press it.

Quick reference

I want to...	Go to	Costs
start from nothing but a topic	Brainstorm	search free, directions paid
see what my papers have in common	Graph	free
know what has already been tried	Novelty check	paid
find open problems	Gaps	free to view
see how the field evolved	Timeline	free
write a cited literature review	Report	plan 1 call, then 1 per question
ask one question	Ask	paid
check my draft against the field	Draft → Analyze	paid
check my citations are real	Citations / <code>refcheck</code>	free
check a source supports my claim	Check citations	paid
avoid a desk reject	<code>check-compliance</code> , <code>check-stats</code>	free
get my bibliography out	<code>export-bib</code>	free

Every claim here reflects behaviour in the shipped tool. Where a signal is a heuristic or an AI judgement, the tool says so on screen — and so does this walkthrough.